Josh Bedwell

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Education

University of California at Berkeley May 2025
M.S. in Molecular Science and Software Engineering | GPA: 3.96 Berkeley, CA

Brigham Young University Apr 2023
B.S. in Computer Science, emphasis in Software Engineering Provo, UT

Experience

Software Engineer, Pallas Security – Santa Clara, CA Jun 2025 – Current

- Develop and maintain production-grade backend services supporting secure agentic-AI
- Contribute to early application and infrastructure architectures
- Design and implement CI/CD pipelines using Docker, uv, GitHub Actions, and AWS services
- Build and manage scalable and cost-effective cloud infrastructure using Terraform
- Integrate over a dozen AWS services for application orchestration, observability and data flow

Research Assistant, BYU STaPL & Chemical Engineering – Provo, UT Jan 2021 – May 2023

- Performed data analysis/NLP for HCI publications on cross cultural interactions and user privacy behavior
- Led prototyping and development of neurodiverse-friendly social platform features
- Refactored, modularized, and tested a new combustion simulation library, enabling publication

Projects

Scenario & File-Based Unit Testing Python Package, PyPi: unittest-scenarios, GitHub 2025

- A framework for platform agnostic state & file based unit testing, useful for data pipeline projects

Phylogenetic Trees & Primers, Transition Biomining & University of British Columbia, GitHub 2025

- Engineered robust data pipelines integrating bioinformatics tools including Alphafold, MMSeqs2
- Extended functionality of TreeSAPP package and improved accessibility for novices

Multimodal Brain Tumor Segmentation Challenge 3D CNN 2024

- Designed 3D convolutional neural networks to predict volumes of different tumor types in 3D MRI images
- Created dynamic analysis tools to reduce input size by over 70% while maintaining >95% input detail
- Designed, trained, and validated models with 7M - 117M params, achieved $r^2 = 0.53$ with exploratory model

Django Hierarchical Models Python Package, PyPi: django-hierarchical-models, GitHub 2024

- A performance-aware Django extension for model schemas such as graph DBs, threads, organizations, etc.

Production ETL Web App, Partners Personnel 2022-2023

- Architected modern Django ETL application to replace local MS Access program
- Designed node-based pipeline architecture, enabling unit testing, serialization, and meeting per-client needs

Publications

CHI '24: Video Interventions to Educate Users about Targeted Advertising on Facebook doi.org
CSCW '23: Comparing Interpersonal Information Disclosure Norms on Twitter doi.org
SoftwareX '23: A Soot Model Library for Combustion Simulation doi.org

Skills

Languages: Python, C, C++, Java, Swift, Rust, JavaScript, Terraform
Machine Learning: TensorFlow, PyTorch, Sklearn
Services: AWS {DynamoDB, API Gateway, Lambda, S3}, GitHub Actions CI/CD
Frameworks/Tools: FastAPI, Django, Flask, Numpy + Pandas, CMake, Docker, SwiftUI